

BECKHAM BENNY ROSS

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EDUCATION

Bachelor of Engineering with Honours in Robotics Systems

Singapore Institute of Technology — September 2024 to September 2028

Diploma in Robotics and Mechatronics

Nanyang Polytechnic — April 2019 to April 2021

Nitec in Aerospace Avionics

Institute of Technical Education, College Central — January 2017 to January 2019

PROFESSIONAL EXPERIENCE

Section Commander (Second sergeant)

Singapore Civil Defence Force — August 2022 – August 2024

- Led and coordinated a team of 4 during operations to ensure swift and safe emergency response
- Ability to Identify fire hazards and assessed risks to prevent potential incidents using systematic protocols
- Ensured compliance with fire safety regulations and maintained readiness of safety systems

Logistic Assistant

VDL Enabling Technologies Group — June 2020 – August 2020

- Assisted with logistics documentation and inventory record management
- Maintained accuracy in supply chain tracking and reported status updates

Maintenance Technician

ST Electronics Training and Simulation — June 2018 – January 2019

- Conducted inventory inspections and reported discrepancies in equipment documentation
- Performed weekly and monthly inspections of training machines to ensure operational readiness
- Assisted in troubleshooting and basic maintenance of equipment and machines

CO-CURRICULAR ACTIVITIES

SIT Project Hub | Student Coach

Singapore Institute of Technology — September 2024 – April 2026

- Guide and mentor staff and students in safe and effective use of equipment and tools
- Support laboratory operations by providing technical and administrative assistance
- Assist in troubleshooting technical issues and resolving problems efficiently

Student Motivator

NYP Primer, Nanyang Polytechnic — April 2019 – April 2021

- Represented NYP as a student motivator, engaging and energising audiences during Singapore National Day events.
- Participated in The Boys' Brigade Share-a-Gift community outreach initiative, supporting efforts to bring assistance and encouragement to beneficiaries.

Air Scout | Rover

Air Scouts of Singapore — January 2017 – January 2019

- Represented ITE at community-based external performances and events, teaching the public basic aerodynamics.
- Represented Air Scouts at the Singapore Airshow 2018

RECENT PROJECTS

Systems Engineering Project 1

- Applied a systems engineering approach using ROS 1 Melodic to develop an autonomous navigation system for the LIMO robot.
- Designed and built a custom testing arena and configured ROS packages for mapping, localisation, and path planning.
- Tested and validated the robot's autonomous navigation performance within the custom environment.

Project 4

- Solved a real-world problem statement provided by CEVA Logistics while considering budget, sustainability, and social impact.
- Developed a computer vision system using RGB and depth cameras for real-time object detection and dimension estimation.
- Designed and calibrated a pipeline to automatically measure package dimensions without manual intervention.
- Delivered a practical solution for logistics and warehouse automation by applying depth sensing, point cloud processing, and object detection.

TECHNICAL SKILLS

3D Modelling & CAD

SolidWorks, AutoCAD Inventor

Prototyping & Hardware

3D Printing, Raspberry Pi, STM32, Mechanical fabrication

Programming & Robotics

Python, C, C++, Spin, ROS/ROS 2, Assembly language

Systems Engineering

Project Management